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REVIEW

Plant-based diets and sports performance: a clinical review

Stephen P. Maier II[1] · Harry M. Lightsey IV[1] · Michael D. Galetta[2] · Andrew O. Usoro[3] · Luke S. Oh[1,4]

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Abstract

Purpose Plant-based diets have become increasingly popular in Western culture. Although studies have examined physiologic health improvements of plant-based diets, there is little data on plant-based diets as it relates to sports performance.

Methods Clinical review of systematic reviews, randomized trials, prospective and retrospective cohort studies available in English on PubMed and Google Scholar databases utilizing combinations of the search terms "athlete, sport, nutrition, diet, vegan, vegetarian, strength, endurance, health, performance, and exercise."

Results There are no significant differences in athletic performance in participants maintaining plant-based diets compared to those consuming omnivorous diets specifically related to strength, power, and endurance. Plant-based diets are at higher risk of predisposing individuals to certain micronutrient deficiencies, reduced protein intake, and lower serum creatine and sex testosterone levels, though supplementation may effectively replace the lacking components. Non-plant-based diets are associated with a higher risk of obesity, type II diabetes mellitus, certain malignancies, and cardiovascular disease, though several studies demonstrate that a particular meat- and fish-filled diet may reduce body weight, blood pressure, fat composition, and all-cause mortality.

Conclusion Though plant-based and mixed omnivorous eaters may vary in their macronutrient and micronutrient intake, disease propensities, and oxygen consumption during exertion, a plant-based diet does not translate into a significantly different physical performance compared to a non-plant-based diet across measures of strength, power, and aerobic/anaerobic performance. For athletes, trainers, nutritionists and dieticians, and physicians, the most important recommendation is to maintain a nutritionally sufficient diet that provides the appropriate levels of vital nutrients and vitamins.

Keywords Plant-based · Diet · Performance · Strength · Health

Introduction

Over the past several decades, there has been increasing interest in the effect of diet on sports performance [1–5]. Since the 1980s, the plant-based diet has been a topic of interest for many physicians, athletic trainers, nutritionists, and athletes alike, with a particular increase in attention and popularity in more recent years [2]. While these diets aim to minimize animal products, variations are subdivided based on the amounts and types of animal-derived products included (Fig. 1). Accompanying this interest, there has been a concordant surge in research efforts to characterize various aspects of performance—including cardiorespiratory fitness, exercise capacity, muscle function, and overall physical performance—and health, as they relate to plant-based diets [1, 2, 4–10].

While a variety of literature exists examining plant-based diets in relation to overall physiologic health, there is little consensus data regarding the efficacy of plant-based diets in relation to sports performance [11]. Proponents of plant-based diets cite the potential for increased carbohydrate and antioxidant intake, as well as the overall alkaline content of plant-based foods, as carrying possible health and

✉ Stephen P. Maier II
spmaier@partners.org

1 Division of Sports Medicine, Department of Orthopaedic Surgery, Sports Medicine Center, Massachusetts General Hospital, 175 Cambridge Street, 4th Floor, Boston, MA 02114, USA

2 Case Western Reserve University School of Medicine, Cleveland, OH 44106, USA

3 Division of Sports Medicine, Cedars-Sinai Marina del Rey Hospital, Marina del Rey, CA 90292, USA

4 Division of Sports Medicine, AdventHealth Orthopedic Institute of Rothman Orthopaedic Institute, Orlando, FL 32805, USA

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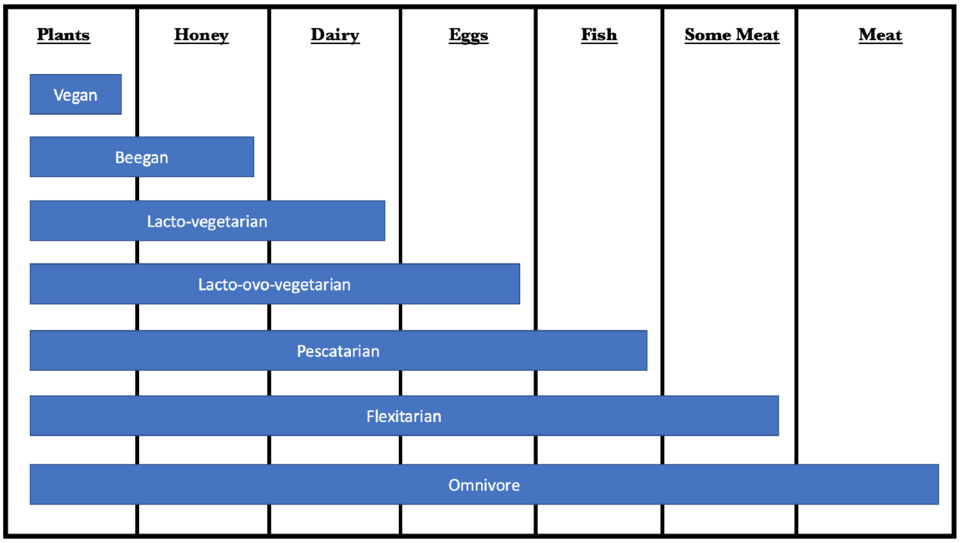

Fig. 1 Graphical depiction of dieting styles. Variations of the plant-based diet include vegan (strict abstinence from consumption of animal-derived food products), beegan (vegan diet with inclusion of honey), lacto-vegetarian (vegan diet that permits inclusion of dairy products), lacto-ovo vegetarian (vegan diet that permits inclusion of dairy products or eggs), pescatarian (vegan diet that permits inclusion of dairy products or eggs, and only meat consumed is fish), and flexitarian (diet predominantly consists of plant-based products with small allowance of meat in the diet). Plant-based diets ubiquitously consist of a high proportion of fruits, vegetables, grains, nuts, seeds, and legumes

performance advantages. Alternatively, critics note deficiencies in micronutrients, reductions in muscle creatine concentration, and potential decreases in serum testosterone associated with plant-based diets and emphasize the implications these drawbacks may have on sports performance. Despite the lack of evidence-based consensus or standard recommendations, several high-performance athletes are using plant-based diets. A recent survey conducted amongst participants at the 2010 Commonwealth games found that roughly 8% of international athletes reported eating plant-based diets, and of those athletes roughly 1% were vegan [12].

In this context, the goal of this clinical review is to consolidate the current literature on plant-based diets in relation to athletic performance. The content and conclusions are intended to be generalizable to physicians, athletic trainers, nutritionists, and athletes in an effort to improve knowledge regarding an increasingly relevant topic. Ultimately, we hope to provide clarity to the existing data and offer evidence-based recommendations related to diet and performance that may be useful to all involved in sport, regardless of level or type of participation .

Methods

This clinical review was conducted utilizing a comprehensive electronic search for published literature in the PubMed database (PubMed.org) and the Ovid MEDLINE libraries of biomedical literature, as well as the Google Scholar search engine between April 2020 and December 2022. Queries were performed by the first author and consisted of scoping searches of English-language articles primarily reporting on plant-based diets and athletic performance. The search strategy utilized varying combinations of the following terms, "athlete, sport, nutrition, diet, vegan, vegetarian, strength, endurance, health, performance, and exercise." No additional search filters were applied. Search results were narrowed following a review of the abstracts by the first author; articles were excluded if the content: (1) Did not pertain to nutrition in the context of athletic performance or exercise, or (2) Did not contribute to specific review subsections. Initial queries yielded 232 studies; this was narrowed to 36 sources following review of abstracts and application of the above exclusion criteria.

Results

Physiologic advantages of plant-based diets

Though once commonly believed that plant-based diets were linked with lower athletic performance, several theories have been proposed to support the notion that plant-based diets are associated with improved performance. It is hypothesized that plant-based diets promote greater carbohydrate intake than non-plant-based diets, which may lead to augmented glycogen stores and therefore a greater supply of energy [13]. It has been demonstrated that carbohydrate ingestion

throughout resistance exercise with or without supplemental protein may promote euglycemia and higher glycogen stores, ameliorate muscle damage, and facilitate improved training adaptations [14].

Niemann hypothesized that the increase in phytochemical and antioxidant composition offered by a plant-based diet may reduce oxidative stress and inflammation and offer improved immunity, with the potential to translate into improved endurance performance [15]. However, this has not been substantiated in the literature, as evidenced by Richter et al. [16].

The elevated alkaline content of plant-based foods has been proposed to reduce intramuscular acidity related to activity, shorten recovery time, and improve performance; however, Hietavala found no overall difference in acid–base statuses—i.e. venous blood pH, strong ion difference, total concentration of weak acids—between plant-based and non-plant-based diets [7].

It has also been theorized that plant-based diets comparatively reduce the viscosity of blood—due to a lower baseline hematocrit born from lower iron availability—causing an increase in tissue oxygenation, which may lead to improved athletic performance [10, 17]. However, as evidenced by Lenz et al., hematologic studies have demonstrated that under physiological conditions when autoregulation is in effect, whole blood viscosity is negligible as a determinant of microvascular perfusion, which serves as a surrogate for oxygen delivery to tissue [18].

Physiologic disadvantages of plant-based diets

There has historically been concern regarding plant-based diets in the athlete, specifically concerning decreased micronutrient availability for high intensity athletes leading to submaximal performance. Fuhrman showed that adherence to a plant-based diet led to a higher risk of micronutrient deficiencies in athletes—most commonly affecting levels of zinc, iodine, vitamin B12, omega-3-fatty acids (DHA and EPA), vitamin D, and taurine [19]. Deficiencies in particular micronutrients are thought to be caused by a reduction in their bioavailability due to greater intake of fiber, as is commonly observed in plant-based dieting [9]. Deficiencies are also likely to stem from lower gross intakes of micronutrients, as plant-based diets exclude some micronutrient rich foods, e.g. red meat (iron, selenium, vitamins A + B12), poultry (zinc, vitamin B12), fish (iodine, omega-3-fatty acids, vitamin D).

Plant-based dieting has also been associated with a reduction in muscle creatine concentrations, and an increase in non-heme iron—found in plant products—which contains iron inhibitors, e.g. calcium, polyphenols, phytic acid [20, 21]. Phytic acid is known to impair or block the intestinal absorption of vital minerals—e.g. zinc, iron, magnesium, calcium—ultimately leading to unabsorbed excretion[22, 23]. Non-heme iron is less bioavailable than heme iron—found in meat, poultry, and fish—with roughly 1–10% of iron absorbed as compared to 25–30% absorption, which too may lead to imbalanced absorption of micronutrients [24].

Furthermore, Raben et al. showed that plant-based dieters exhibited decreases in total fasting serum sex testosterone over the course of a 6-week study period as compared to mixed, meat-rich dieters (13.7 vs. 17.4 nmol/L) [9]. Despite these differences, the authors found no significant difference in endurance performance between diets. Hill similarly found that men fed a vegetarian diet for 2 weeks displayed decreases in their nocturnal release of testosterone, with average plasma testosterone levels dropping from 585 to 401 ng/100 ml. [25]. However, there is evidence that circulating serum testosterone is directly related to muscle mass and strength with a dose-dependent response, and that suppression of elevated circulating testosterone levels in athletes reduces performance—a response that is reversed when suppression ceases [26].

Comparing the plant-based diet and the omnivorous diet

Protein intake

As per the Academy of Nutrition and Dietetics, plant-based diets on average meet or exceed the daily recommended dietary intake for protein when caloric intake is sufficient [27]. Lynch performed a cross-sectional study in female endurance athletes that showed lower total protein intake in plant-based athletes but equivalent protein levels as a function of body weight (grams/kilogram), and the average protein consumption in plant-based and non-plant-based athletes fell within the recommended intake range as recommended by the Academy of Nutrition and Dietetics [28]. In Gibson-Smith's cross-sectional study of young, healthy, active climbers, vegan/vegetarian climbers had significantly lower protein intake than omnivorous climbers (1.25 vs. 1.71 g/kg/day), and climbers who used a protein supplement had significantly higher protein intake than those that did not [29].

Early literature maintained that athletes were recommended to consume protein at levels consistent with those of the general population, which equates to roughly 12–15% of energy from protein [30]; however, more recent research suggests that athletes' protein intake requirements exceed the general population's [11]. Recent indicator amino acid oxidation studies by Bandegan et al. demonstrate athletes' protein requirements are around 1.9 g/kg of bodyweight [31, 32]; thus, plant-based diets risk falling short of optimal levels

of protein intake for athletes. For athletes that do not reach optimal levels of protein intake, protein supplementation has been shown to increase strength and muscle mass and may provide a necessary adjunct to dietary protein [33].

Micronutrients

Plant-based dieting has been shown to increase risk of micronutrient deficiencies in athletes, as noted previously, prompting recommendations for supplementation [19]. The recommended dietary intake of zinc, iodine, and omega-3-fatty acids have been shown to be roughly equivalent in plant-based eaters and non-plant-based eaters [5, 12]. However, some research has shown that vegetarians often exhibit lower than recommended zinc levels [34], and males with low zinc levels have been shown to have lower testosterone levels [35].

Cobalamin is found strictly in animal-based products (e.g. fish, meat, eggs, poultry, dairy), and its recommended daily intake increases with age. The Academy of Nutrition on Vegetarianism recommends all plant-based dieters should supplement their diets with Vitamin B12 to avoid deficiency, with strong recommendations to vegan dieters [12, 36, 37]. Similarly, retinol—also known as Vitamin A, critical for vision, immune function, growth, cell division, and development—is only found in animal-based products [38]. Beta carotene, a nutrient that may be converted to retinol once ingested, is found in non-animal-based products such as green leafy vegetables; however, beta-carotene to retinol conversion is often unreliable [39]. Furthermore, carnitine—a compound with critical function at the cellular level in extracting energy from fatty acids and maintaining mitochondrial homeostasis during fatty acid oxidation—is consumed primarily through animal-based products, as plant-based products contain trace amounts of carnitine, and has been shown to enhance athletic performance [40, 41].

Omega-3-fatty acids – essential nutrients that cannot be made by the body and must therefore be ingested through diet—can be divided into three classes: alpha-linolenic acid (ALA), eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA). A major scientific question that continues to be investigated is whether the physiologically beneficial effects of ALA occur before or after ALA is biologically converted into EPA and DHA. ALA is primarily found in plant-based sources, including flaxseed, chia seed, perilla, walnut, rapeseed, and soybean oils; whereas EPA and DHA are primarily derived from non-plant-based sources, specifically seafood and fish oils [42].

Furthermore, though plant-based dieters tend to have lower iron stores compared to omnivores, data suggests that bioavailability of non-heme iron increases over time to adapt to increased non-heme iron intake, which is predominantly found in plant-based diets [10, 43]. Vitamin D is recommended if dietary contributions are minimal and in those following plant-based diets, despite lower bone mineral densities on average [44]. In a large-scale cross-sectional study in the United Kingdom of over 35,000 subjects adjusted for age, sex, and non-dietary factors, Appleby found no meaningful increase in fracture risk in vegetarians (incidence rate ratio: 1.0) but an increased fracture risk in vegans (incidence rate ratio: 1.30) when compared to meat eaters [45]. However, in a population study of approximately 65,000 participants, Tong found that when compared to meat eaters, the risk of hip fracture is increased in pescatarians, vegetarians, and vegans by 1.26, 1.25, and 2.31 times, respectively [46].

Creatine

Systemic creatine levels—which are associated with muscular strength and power—have been found to be lower in those maintaining plant-based diets as compared to non-plant-based dieters [4]. Blancquaert et al. found that omnivorous women who changed to a vegetarian diet experienced reductions in their plasma and muscle creatine levels over 3 and 6 months; notably, these decreases were ameliorated with low-dose creatine supplementation [47]. In a comprehensive study by Burke et al., the effect of creatine on weight training and performance was compared between plant-based dieters and non-plant-based dieters, which showed that baseline intramuscular concentrations of total creatine were significantly lower in plant-based diets [20]. The study also demonstrated that creatine supplementation resulted in a 19% increase in one-repetition maximum bench press compared with an 11% increase for placebo supplementation with no difference between diets; and a significant positive correlation was noted between change in intramuscular creatine concentration and change in bench press ($r = 0.62$) and leg press ($r = 0.52$) with creatine supplementation, highlighting the contribution of creatine to physical strength. Plant-based diets supplemented by creatine were associated with significantly greater increases in muscle concentrations of phosphocreatine and total creatine, lean tissue mass, and total work output when compared to non-plant-based diets similarly supplemented by creatine, suggesting a greater deficit in baseline creatine in plant-based dieters. This data suggests that creatine may be a helpful supplement in athletes who follow a plant-based diet.

Immunity

Although injury is the most likely reason to sideline an athlete, infection or illness may play a substantial role and pose a potentially serious problem for athletes as well. There has been described a bidirectional relationship between nutrition and infection; namely that poor nutritional status leads to depressed immunity and heightened risk of infection, and infection leads to greater levels of required

energy by the host which paradoxically diminishes appetite and reduces levels of absorbed nutrients [5]. Some research has focused on energy availability as a surrogate for nutrition, which is assessed by a constellation of measures (e.g. injury prevalence, gastrointestinal symptoms, menstrual details, orthostatic hypotension). Drew found that energy availability was inversely related to infection risk in female athletes preparing for the Summer Olympic games of 2016 in Rio de Janiero, Brazil, with low energy availability yielding a four to eight times higher risk of developing an upper respiratory infection during intense pre-Olympic training [48, 49]. Conversely, however, Richter conducted a crossover design study in eight elite endurance trained male athletes who maintained a plant-based diet and a non-plant-based diet for 6 weeks each with a 4-week interim washout period, and found no significant differences in concentration or composition of mononuclear cells (CD3+, CD4+, CD8+, CD14+, and CD16+), nor in the activity of natural killer cells between groups [16].

Health

Plant-based diets are associated with lower body mass index (BMI) when compared to non-plant-based diets. Across multiple studies, the average BMI is lower in plant-based eaters as compared to non-plant-based eaters – 22.5–23.6 vs. 24.4–28.8, respectively [50, 51]. Using a different metric, the prevalence of overweight or obese individuals is on average lower in plant-based eaters as compared to non-plant-based eaters – 25% vs. 40% [52].

Plant-based eaters have also been shown to be less likely to develop type II diabetes mellitus (T2DM). In a survey study of over 60,000 participants, Tonstad et al. showed that the prevalence of T2DM in non-plant-based eaters was greater than twice the prevalence in plant-based eaters [50]. In a later study with a similarly large-scale sample, Tonstad et al. found that plant-based diets were associated with a 54% risk reduction of developing T2DM when compared to non-plant-based diets and controlling subjects for age; vegan diets had a 77% risk reduction. The trends remained similar and strong when the samples were controlled for BMI [37].

According to the Academy of Nutrition and Dietetics, plant-based diets are associated with an overall reduced cancer risk—particularly gastrointestinal cancer—when compared to non-plant-based diets [27]; however, further analysis showed no significant difference in cancer-related mortality between groups [53]. Several studies have found a lower overall incidence of malignancy as well as a significant risk reduction in certain types of cancer in plant-based diets when compared to non-plant-based diets, however, it is unclear if this effect is dependent on confounding factors [54, 55]. Fruits and vegetables contain phytochemicals (e.g. phytosterols, salicylates, and fiber) which have been shown to interfere with immune cell regulation, proinflammatory cytokine synthesis, and gene expression, thereby regulating immunoresponse to inflammation as evidenced by lowered serum IL-6 and C-reactive protein levels [56–59]. This may reduce the risk or prevent the progression of chronic diseases such as cardiovascular disease, T2DM, and some cancers [58]. Bingham et al. showed that dietary fiber intake may be related to development of colorectal cancer, as high fiber diets confer a 25% relative risk reduction when compared to low fiber diets [60]. Conversely, the World Cancer Research Fund and American Institute for Cancer Research noted a positive correlation between consumption of processed and red meat and risk of colorectal cancer—i.e. 100 g/day of red meat and 50 g/day of processed meat increased the risk of colorectal cancer by 18–29% and 18–21%, respectively [61, 62].

Plant-based diets have been associated with reduced intake of total and saturated fats, weight, lower cholesterol levels, and reduced incidence of cardiovascular disease [53, 54, 63, 64]. In a meta-analysis of 11 randomized controlled trials comparing mixed diets to plant-based diets for an average duration of intervention of 24 weeks, Wang et al. found significant reductions in total cholesterol, low-density lipoprotein cholesterol, and high-density lipoprotein cholesterol, with an associated 10% risk reduction in development of heart disease [65]. Orlich et al. studied over 73,000 subjects and found that plant-based diets were associated with a 13% decrease in risk of developing cardiovascular disease and a 19% risk in development of ischemic heart disease [53], and Crowe et al. studied nearly 45,000 subjects in a United Kingdom-based registry study and found that plant-based diets were associated with a 32% reduction in risk of hospitalization or mortality over the 12-year study period [63]. Multiple large-scale studies have demonstrated that plant-based diets are associated with lower mean systolic and diastolic blood pressure as compared to non-plant-based diets, and strictly vegan diets have the lowest average blood pressure and overall incidence of hypertension [36, 66, 67].

Several studies have shown that a predominantly meat and fish-filled diet devoid of processed foods, dairy products, and refined grains–referred to as the Paleolithic diet–have beneficial health outcomes. In a healthy cohort of 14 volunteers who consumed a Paleolithic diet for 3 weeks, Österdahl et al. found decreases in mean weight, BMI, waist circumference, systolic blood pressure, and fat composition as compared to their pre-intervention baselines, with an improvement in antioxidant levels and a decrease in energy intake by roughly one-third of their prior [68]. Whalen et al. studied over 13,000 subjects for a median time of over 6 years and found that Paleolithic dieting principles were inversely proportional to all-cause and cause-specific mortality [69]. In a randomized cross-over study over 3 months of 13 patients with T2DM, Jönsson et al. found lower mean values of HbA1c, triacylglycerol, diastolic

blood pressure, weight, BMI, and waist circumference, and higher mean values of high density lipoprotein cholesterol in patients consuming a Paleolithic diet as compared to a diabetic diet as prescribed by the American Diabetes Association (ADA) [70]. Frassetto et al. published similar results with comparatively better improvements in glycemic levels and lipid levels in diabetic patients consuming Paleolithic diets as compared to ADA diets [71].

It is important to recognize the limitation of healthy user bias in the aforementioned observational studies, which may restrict their external validities. Healthy user bias, or the healthy volunteer effect, refers to a bias created by the participants that enroll in epidemiologic studies, who are likely to be healthier than the general population. This limitation is difficult to overcome without intervention studies.

Performance outcomes

Multiple studies have been conducted to investigate how performance relates to diet (Table 1). In short, a recent review conducted by Craddock found no differences in exercise performance across multiple activity types and performance measures when comparing plant-based to non-plant-based diets [2]. Shaw similarly conducted a review, in which the authors concluded that a plant-based diet neither hinders nor aids in exercise performance and were in favor of consultation with dietary professionals with a sports background to guide safe and optimal plant-based dieting in the athlete [72].

Strength

Several research efforts have measured overall strength gain following a 12-week resistance training program, which consistently demonstrate no significant differences in performance measures (i.e. one-repetition maximum weight lifted, maximum repetitions of weight lifted to failure) between those randomized to a plant-based diet and those randomized to a non-plant-based diet, though Wells noted a significantly greater strength increase in isolated knee extension exercises in the plant-based diet group, and Campbell found decreased fat-free mass and whole-body muscle mass in the plant-based diet group [6, 10, 73]. Lynch found no significant differences in peak torque during leg extensions and flexion exercises using an isokinetic dynamometer between plant-based and omnivorous endurance athletes [28]. Veleba conducted a 12-week randomized controlled trial of seventy-four participants with T2DM who were placed on a calorie-restricted diet (randomly assigned to conventional diet vs. vegetarian diet) combined with an aerobic exercise program performed under professional supervision, which found that maximal performance measured in Watts significantly increased over the study period (21% increase), an increase that was not observed in the conventional diet group [74].

In a 5-week randomized control trial consisting of a sprint training program comparing plant-based diets (with creatine supplementation to compensate for the known deficits in plant-based diets) to non-plant-based diets, Baguet found no significant differences in power output while sprinting or changes in muscle carnosine concentrations from baseline, though the plant-based eaters had a lower composition of protein (13.1% vs. 15.8%, respectively) and higher composition of fat (28.0% vs. 26.0%, respectively) in their diets [75]. Page found no significant differences in vastus lateralis thickness (26.1 mm vs. 27.8 mm) or maximal voluntary isometric contraction (627 N vs. 551 N), between habitual vegans and habitual omnivores, respectively, in a cross-sectional study of twenty-six young, healthy, recreationally active men [76]. In a cross-sectional study of thirty-eight young, healthy men who participated in a 12-week professionally supervised resistance training program, Hevia-Larraín similarly found that habitual vegans and habitual omnivores supplemented to 1.6 g/kg/day with soy or whey protein, respectively, sustained significant increases over the study period in leg lean mass (1.2 kg, 1.2 kg), rectus femoris cross-sectional area (1.0 cm^2, 0.9 cm^2), vastus lateralis cross-sectional area (2.2 cm^2, 2.8 cm^2), vastus lateralis muscle fiber type I cross-sectional area (741 μm^2, 677 μm^2), and type II cross-sectional area (921 μm^2, 844 μm^2), and leg-press one-repetition maximum (97 kg, 117 kg), without significant differences between groups [77].

Endurance

Similar to strength evaluation across diet types, there has been no significant measurable difference in endurance between plant-based and non-plant-based performers. Raben conducted a randomized control trial with a crossover design and interim washout period comparing physiologic and performance outcomes in elite endurance athletes who maintained a plant-based diet versus a non-plant-based diet for six weeks of each, and found a small decrease in testosterone during the plant-based diet phase, but no significant difference in maximal oxygen uptake, time to exhaustion, maximal voluntary contraction, or isometric endurance, which were used as surrogates for exercise endurance capacity [9].

In a crossover designed study of nine recreationally active men who consumed a plant-based diet and a non-plant-based diet for 4 days each with a 16-day interim wash-out period, Hietavala demonstrated exercise economy was poorer in plant-based diets compared to omnivorous diets as the oxygen consumption was significantly higher (~13%) at submaximal stages of cycle ergometer testing (40%, 60%,

Table 1 Summary of key findings from studies included in clinical review

Study	Study Type & LOE	Sample	Methods	Findings
Baguet 2011 [75]	Intervention Study Level 1	Healthy students – 10 omnivorous diet (average age 21.5 yr, 5 M/5F), 9 lacto-ovo vegetarian diet (average age 20.8 yr, 5 M/4F)	Randomized to omnivorous vs. lacto-ovo vegetarian diet while completing 5-wk sprint training program (cycling + running)—all participants were supplemented with creatine (1 g/day)	No significant differences in power output while sprinting or changes in muscle carnosine concentrations from baseline, but vegetarian group had lower average protein composition (13.1% vs. 15.8%, respectively) and higher average fat composition (28.0% vs. 26.0%, respectively) in diet. Carnosine content of soleus was found increased (+11%) in the omnivorous group and decreased (−9%) in the vegetarian group, though no other muscle groups demonstrated this finding
Bandegan 2019 [32]	Cross-sectional Study Level 3	Healthy young endurance-trained males, ($N=8$, average age: 26.6 yr)	Cross-sectional study; measured estimated average requirement of protein	Found to be 2.1 g/kg/day, which exceeds the estimated average requirement of protein as stated by the institutes of medicine by roughly 3.5 times
Bandegan 2017 [31]	Cross-sectional Study Level 3	Healthy young male bodybuilders, ($N=8$, average age: 22.5 yr)	Cross-sectional study; measured estimated average requirement of protein on a non-training day	Found to be 1.7 g/kg/day, which exceeds the estimated average requirement of protein as stated by the institutes of medicine by roughly 2.6 times
Benvenga 2005 [41]	Clinical Review Level 2	N/A	Clinical Review	High-intensity exercise (e.g. endurance training) raises muscular L-carnitine demands and lowers muscular L-carnitine composition. Supplementation of L-carnitine lowers the muscular depletion rate of glycogen and increases fatty acid oxidation in muscle, thus improving exercise performance by a proposed mechanism of increased proliferation of capillaries in skeletal muscles induced by training. L-carnitine supplementation benefits persons who remain active and stimulate skeletal muscle, thus the role for supplementation in sedentary persons is limited

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Blancquaert 2018 [47]	Intervention Study Level 1	Omnivorous women ($N=40$)	Participants randomly assigned to one of three dietary groups: omnivorous diet ($N=10$), vegetarian diet without supplementation ($N=15$), vegetarian diet + daily β-alanine & creatine supplementation (0.4–0.8 g/day, 1 g/day; respectively; $N=15$), for six-month trial	In the non-supplemented vegetarian group, plasma creatine significantly decreased from 0 to 3 months and 0 to 6 months, whereas in the vegetarian group with supplementation plasma creatine significantly increased during the same timeframe. The non-supplemented vegetarian group also demonstrated significantly reduced muscle creatine levels at 3 months. Muscle carnosine content significantly increased in the vegetarian group with supplementation from 0 to 3 months and 0 to 6 months, while muscle carnosine content did not significantly change in the omnivorous or non-supplemented vegetarian groups. There were no significant changes in plasma or muscle carnitine levels in any of the groups, nor significant changes in performance measures (i.e. VO2 max, cycling time to exhaustion) in any of the groups
Boutros 2020 [78]	Cross-sectional Study Level 3	Healthy, young physically active women (average age: 25.6 yr)	Cross-sectional survey of 28 vegan dieting and 28 omnivorous dieting women for two consecutive years	No significant differences in percent body fat, BMI, physical activity level, muscle strength between groups. Vegan group had a significantly higher VO2 max (44.5 vs. 41.6 ml/kg/min) and submaximal endurance time to exhaustion (12.2 vs. 8.8 min) than omnivorous group

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Burke 2003 [20]	Intervention Study Level 1	Vegetarian ($N=18$) and non-vegetarian ($N=24$) participants aged 19–55 yr	Participants were randomly sorted into groups with and without creatine supplementation and participated in the same 8-week resistance training program	Baseline intramuscular concentrations of total creatine were significantly lower in vegetarian participants. Creatine supplementation was found to have in a 19% increase in one-repetition maximum bench press compared with an 11% increase for placebo supplementation, with no difference found between diets. Change in intramuscular creatine concentration was positively correlated with change in bench press ($r=0.62$) and leg press ($r=0.52$) with creatine supplementation. Protein supplementation in vegetarians significantly increased muscle concentrations of phosphocreatine and total creatine, lean tissue mass, and total work output when compared to non-vegetarians similarly supplemented by creatine
Campbell 1999 [73]	Intervention Study Level 1	Male participants (age range: 51–69 years)	Participants were randomly assigned to continue standard omnivorous diet ($N=9$) vs. consume a lacto-ovo vegetarian diet ($N=10$) while completing a resistance training program for 12 weeks	Maximal strength increased in muscle groups of upper and lower body trained during the study period (range: 10–38%), independent of diet. Whole-body density, fat-free mass, and whole-body muscle mass increased in omnivorous group but decreased in vegetarian group. Vastus lateralis type II muscle fiber cross-sectional area increased over the study period independent of diet, but increase was found to be greater in the omnivorous group (16.2%) than in the vegetarian group (7.3%)
Craddock 2016 [2]	Systematic Review Level 1	Eight articles	Systematic Review	No consistent significant differences in strength, power, aerobic/anaerobic performance, and immune function between study participants that consumed a plant-based diet vs. a non-selective diet

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Davey 2021 [82]	Case Study Level 5	25-year-old rugby football player transitioning from an omnivorous diet to a vegan diet	Case Study	Lower intake of protein and fat, and a higher intake of fiber and long-chain poly-unsaturated fatty acids with new diet. Levels of calcium, iodine, vitamin D, and vitamin B12 were on lower limit of adequate intake, prompting supplementation. Lean body mass increased after adopting the vegan diet compared to prior seasons of omnivorous eating, without significant changes in body fat percentage. No significant differences observed when comparing the pre-vegan and post-vegan seasons in any parameter of performance (i.e. total distance run, accelerations, decelerations, repeated high-intensity efforts, high speed running distance, conditioning distance)
Devries 2014 [84]	Meta-analysis Level 1	Adults (> 45 yr old)	Meta-analysis ($N = 12$ studies) of pooled participants ($N = 357$) evaluating strength differences between resistance training alone vs. resistance training + creatine supplementation	Creatine-supplemented resistance training associated with increases in one-repetition maximum chest press, one-repetition leg press, and 30-s chair stand test in older individuals, when compared to non-supplemented resistance training
Devrim-Lanpir 2021 [90]	Cross-sectional Study Level 3	Ultra-marathonists	Cross-sectional study of 35 vegan vs. 35 omnivorous athletes participating in a meta-genomic study to evaluate the effects of diet type on metabolic processes including redox homeostasis, muscle fatigue, energy availability, and gut microbiome adaptations in extreme endurance athletes. Bodily specimens, laboratory studies, food logs, and questionnaires are included in the methodology of this study	Results not yet available for review

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Drew 2017 [48]	Survey Study Level 4	Olympic athletes (47 male, average age: 25.8 yr old; 85 female, average age 24.3 yr old) from 11 sports	Survey study of Olympic athletes ($N=132$) three months prior to participating in the 2016 Summer Olympics in Rio de Janeiro	Athletes with low energy availability had an odds ratio of 3.8 times greater risk of sustaining an upper respiratory illness in the prior month. Poor sleep quality and daytime drowsiness were not associated with upper respiratory illness. The two most significant factors related to illness development were stress and anxiety (as measured by (1) Depression, Anxiety, and Stress Questionnaire, (2) Recovery-Stress Questionnaire)
Drew 2017 [49]	Cross-sectional Study Level 3	Olympic athletes (26 male, 55 female) from 11 sports	Cross-sectional survey study of athletes nine months prior to participating in the 2016 Summer Olympics in Rio de Janeiro comparing illness cases and non-ill controls	Odds ratios associated with illness were significantly related to low energy availability (OR = 7.4), depression symptoms (OR = 8.4) and females (OR 9.4)
Durkalec-Michalski 2022 [79]	Intervention Study Level 1	Adult moderately-trained CrossFit participants ($N=20$, 8 M/12F)	Participants were randomly assigned to a customary mixed diet ($N=10$, 4 M/6F, average age: 30.5 yr) or a vegetarian diet ($N=10$, 4 M/6F, average age: 31.0) while performing the 4-week high-intensity functional training study protocol	There was a significant increase in the dietary fiber intake in the group that adopted the vegetarian diet for 4wk (59.7 vs. 44.9 g/day). At the conclusion of the study period, carbohydrate and dietary fiber intake were significantly greater in the vegetarian group than the mixed diet group, and fat intake was significantly lower in the vegetarian group compared to the mixed diet group. Regarding performance, the mixed diet group showed significant improvements in squat + Wall Ball repetitions over the study period, and the vegetarian group showed significant improvements in deadlift + Wall Ball + rowing repetitions as well as the + FightGoneBad performance simulation. No significant differences were identified in or between groups for blood biochemical indices
Fuhrman 2019 [19]	Clinical Review Level 2	N/A	Clinical Review	Plant-based dieting has benefits to immunologic and cardiovascular health, however, there is a higher risk of micronutrient deficiencies in athletes adhering to such diets—most commonly affecting levels of zinc, iodine, vitamin B12, omega-3-fatty acids (DHA and EPA), vitamin D, and taurine

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Gibson-Smith 2020 [29]	Cross-sectional Study Level 3	Experienced elite climbers ($N=40$, 20 M/20F, average years experience: 8.8, average age: 30.3 yr)	Cross-sectional study of nutritional assessment of participants	Found that 27.5% were vegan/vegetarian (6 M/5F). Of the total cohort, 30% of females met criteria for iron deficiency—mean serum ferritin was significantly lower in vegan/vegetarians than omnivores (33.2 vs. 57.5 µg/L) and in females than males (51.4 vs. 102.7 µg/L). Vegan/vegetarian climbers had significantly lower protein intake than omnivorous climbers (1.25 vs. 1.71 g/kg/day). Climbers that used a protein supplement had significantly higher protein intake than those that did not, and there was no significant difference in protein supplementation between vegans/vegetarians and omnivores (36.4% vs. 35.9%). No significant difference in protein intake or iron intake between vegans and vegetarians
Handelsman 2018 [26]	Clinical Review Level 2	N/A	Clinical Review	Circulating testosterone levels are greater in men than women. Both sexes demonstrate dose–response relationships between circulating testosterone and muscle mass and strength. Suppression of circulating testosterone levels results in negative effects on performance, which is reversed when testosterone suppression is discontinued
Haub 2005 [6]	Intervention Study Level 1	Healthy men ($N=21$, average age: 65 yr)	Participants consumed a a lacto-ovo vegetarian diet while completing a resistance training program for 12 weeks. Randomly assigned 10 participants to receive 0.6 g-protein/kg per day from portioned quantities of beef vs. 11 participants to continue with texturized vegetable protein foods	Found no significant differences in performance measures (i.e. one-repetition maximum weight lifted, maximum repetitions of weight lifted to failure) between groups

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Haub 2002 [89]	Intervention Study Level 1	Healthy men ($N=21$, average age: 65 yr)	Participants consumed a a lacto-ovo vegetarian diet while completing a resistance training program for 12 weeks. Randomly assigned 10 participants to receive 0.6 g-protein/kg per day from portioned quantities of beef vs. 11 participants to continue with texturized vegetable protein foods	Improvements observed in both groups in maximal dynamic strength of all trained muscle groups (14–38%). Cross-sectional muscle area of vastus lateralis increased with resistance training in both groups (lacto-ovo vegetarians: 4.2%, beef diet: 6.0%)—no significant differences between groups. Found no significant differences between or within groups in body composition, resting energy expenditure, and concentrations of muscle creatine/phosphocreatine/total creatine
Hernández-Martínez 2020 [83]	Case Study Level 5	Habitually omnivorous professional powerlifter who maintained a vegan diet for 6 weeks while training	Case Study	Body mass decreased by 2.4% (79.3 to 77.4 kg) and fat mass decreased by 24% (15.0 to 11.4 kg). The athlete's fat free mass increased by 2.6% (64.3 to 66.0 kg). No changes in the athlete's perceived energy availability were noted
Hevia-Larrain 2021 [77]	Cross-sectional Study Level 3	Healthy men	Cross-sectional study of habitually vegan ($N=19$) and habitually omnivorous ($N=19$) who participated in a 12 week professionally supervised resistance training program. Baseline protein intake for vegans and omnivores was supplemented to 1.6 g/kg/day with soy or whey protein, respectively	Significant increases over the study period in leg lean mass (1.2 kg, 1.2 kg), rectus femoris cross-sectional area (1.0 cm2, 0.9 cm^2), vastus lateralis cross-sectional area (2.2 cm2, 2.8 cm^2), vastus lateralis muscle fiber type I cross-sectional area (741 µm2, 677 μm^2), and type II cross-sectional area (921 µm2, 844 μm^2), and leg-press one-repetition maximum (97 kg, 117 kg) in the vegan and omnivorous groups, respectively. There were no significant differences between groups in any of the above measures
Hietavala 2012 [7]	Intervention Study Level 1	Healthy, recreationally active men ($N=9$)	Matched cross-over study; unrestricted mixed diet vs. low-protein vegetarian diet for 4 days of each diet with a 10–14 day interim washout period	Found oxygen consumption was significantly higher in low-protein vegetarian diet (~13%) at submaximal stages of cycle ergometer testing (40%, 60%, and 80% of maximum oxygen capacity). No overall difference in acid–base statuses—i.e. venous blood pH, strong ion difference, total concentration of weak acids—between diet types

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Kerksick 2017 [14]	Review Level 2	N/A	Clinical Review	Carbohydrate ingestion throughout resistance exercise may promote euglycemia and higher glycogen stores, ameliorate muscle damage, and facilitate improved training adaptations (with or without supplemental protein)
Leischik 2014 [81]	Case Study Level 5	48-year-old ultra-triathlete	Case Study	A three-time Ironman (total = 11.4 km swim + 540 km bike + 125 km run) maintains a raw vegan diet and finished in 41 h, 18 min. No differences in his spiroegometry, echocardiography, and basic laboratory results compared to 10 omnivorous ultra-marathonists
Lynch 2016 [28]	Cross-sectional Study Level 3	Elite adult endurance athletes	Cross-sectional study of vegetarian ($N = 27$) and omnivorous ($N = 43$) athletes tested for oxygen uptake and strength	Vegetarian athletes had higher levels of oxygen consumption than omnivorous athletes at similar exercise intensities (53% vs. 47%, respectively). Vegetarians had lower total protein intake than omnivores, however protein intake as a function of body mass did not differ between groups. No difference between groups in peak torque dynamometer testing (strength). No difference in oxygen consumption for between groups, but female vegetarians had higher peak oxygen intake (53.0 vs. 47.1, respectively) than female omnivores
Lynch 2018 [4]	Clinical Review Level 2	N/A	Clinical Review	Review of literature regarding plant-based vs. non-selective omnivorous diets, specifically exploring their nutritional contents and performance effects. Found differences in macro- and micronutrient intake between diet types (e.g. higher total energy, saturated fat, cholesterol, sodium, and protein in omnivores; higher fiber and calcium in plant-based dieters; higher iron ingestion but conversely lower iron stores in plant-based dieters), as well as physiological differences (e.g. lower total body creatine and plasma carnitine among vegetarians). No gross differences in exercise performance were consistently identified between diet types across measures and activity types

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Nebl 2019 [8]	Cross-sectional Study Level 3	Recreational runners with similar baseline metrics (i.e. training frequency, time, running distance)	Cross-sectional study of vegan ($N=24$), lacto-ovo vegetarian ($N=26$), and omnivorous ($N=26$) athletes	No significant differences in maximum power output, arterial lactate throughout exercise, and maximum lactate
Page 2021 [76]	Cross-sectional Study Level 3	Healthy, recreationally active men	Cross-sectional study of habitual vegans ($N=9$) vs. habitual omnivores ($N=16$)	No differences found in vastus lateralis thickness (26.1 mm vs. 27.8 mm), fascicle pennation angle (16.6° vs.17.7°), maximal voluntary isometric contraction (627 N vs.551 N), or V̇O2max (40.8 ml/kg/min vs.35.8 ml/kg/min) between habitual vegans and habitual omnivores, respectively
Pohl 2021 [91]	Clinical Review Level 2	N/A	Clinical Review	Review of fourteen studies investigating the effects of diet (vegan/vegetarian vs. mixed) on exercise performance. Concluded that exercise performance is dependent upon an interplay of multiple physiological subsystems, both directly via uptake of specific nutrients and indirectly via nutrient-induced modulation of the molecular environment. The composition of a plant-based diet may provide an advantage to endurance athletes, as endurance performance relies on skeletal muscle mitochondrial and capillary density, hemoglobin concentration, endothelial function, functional heart morphology and availability of carbohydrates. A plant-based diet may be disadvantageous to strength-based athletes, as strength performance relies on creatine and protein availability for muscle protein synthesis. On a molecular level, exercise performance may be influenced by molecular muscular adaptations and signaling in tissues downstream from dietary-acquired nutrients (i.e. leucine, taurine, DHA, EPA, SCFA). Furthermore, dietary choices affect the gut microbiome, which consequently impacts intestinal inflammation, carbohydrate, protein anabolism, and protein fermentation processes, and fat oxidation; plant-based dieting may lead to a more favorable gut microbiome and thus improved exercise performance

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Raben 1992 [9]	Intervention Study Level 1	Endurance-trained male athletes ($N=8$)	Matched cross-over study; unrestricted mixed diet vs. lacto-ovo vegetarian diet for 6 weeks of each diet with a 4-week interim washout period	Found a small decrease in testosterone during the vegetarian diet phase (13.7 vs. 17.4 nmol/L), but no significant difference in maximal oxygen uptake, time to exhaustion, maximal voluntary contraction, or isometric endurance (surrogates for exercise endurance capacity)
Richter 1991 [16]	Intervention Study Level 1	Elite endurance trained male athletes ($N=8$)	Athletes were assigned to 6 weeks of each diet (meat-rich diet vs. lacto-ovo vegetarian diet) with a 4-week interim washout period	Found no significant differences in concentration or composition of mononuclear cells (CD3+, CD4+, CD8+, CD14+, and CD16+) or activity of natural killer cells between groups
Shaw 2022 [72]	Clinical Review Level 2	N/A	Clinical Review	A plant-based diet may offer favorable health effects—likely secondary to increased intake of whole grains/fruits/vegetables as opposed to avoidance of animal products. Strict plant-based dieting, however, may lead to insufficient levels of macro- and micronutrients. Careful attention should be paid towards monitoring of those nutrients at risk of deficiency. Overall, a wholesome plant-based diet with proper monitoring and attention neither hinders nor aids in athletic performance. The authors recommend consultation with dietary professionals with a sports background to guide safe and optimal plant-based dieting in the athlete

Table 1 (continued)

Study	Study Type & LOE	Sample	Methods	Findings
Veleba 2016 [74]	Intervention Study Level 1	Type II diabetic participants ($N=37$ in vegetarian diet group, $N=37$ in conventional diet group)	Participants randomly assigned to vegetarian vs. conventional diet group for 12 weeks, during which the diets were calorie-restricted (500 kcal/day), and combined with an aerobic exercise program performed under professional supervision	The vegetarian group exhibited increases in maximal performance (21% improvement, Watt max) and VO2max (12% increase), whereas the conventional diet group did not have significant changes in these measures. Resting energy expenditure significantly decreased in the conventional diet group and did not significantly change in the vegetarian group. Fasting oxidation of carbohydrates decreased by 89% in the conventional group but did not significantly change in the vegetarian group. There were no significant changes in respiratory quotient, fasting oxidation of fats, or fasting oxidation of proteins in either group
Wells 2003 [10]	Intervention Study Level 1	Healthy men ($N=21$, age range: 59–78 years)	Participants completed a resistance training program for 12 weeks, 10 subjects were randomized to a beef diet and 11 subjects were randomized to a lacto-ovo vegetarian diet	Found no significant difference in total iron intake between groups; but the vegetarian group had 3–4 × less intake of bioavailable iron than the beef group. No significant differences in serum iron, total iron binding capacity, transferrin saturation, and transferrin receptor between groups. Maximal strength increased in all muscles trained during the study period (range: 14–38%), no significant difference in strength increase between groups in leg press, seated leg curl, chest press, and arm pull; however, there was a significantly greater increase in strength for the knee extension exercise in the vegetarian group
Wirnitzer 2014 [80]	Case Study Level 5	30-year-old vegan mountain biker during the Transalp Challenge in 2004	Case Study	Accomplishments included: 22,500 m altitude climbed, 662 km completed in total distance, finished the 8-stage event in 42 h (rank 16th in mixed race group—514 min behind group winner, 4.6 W/kg relative peak power output, average heart rate was 79.5% of laboratory-determined maximum. In reference to diet, 35% of calories came from carbohydrates

and 80% of maximum oxygen capacity) [7]. This result was corroborated in female endurance athletes by Lynch et al., who showed plant-based athletes had higher levels of oxygen consumption than omnivores at similar exercise intensities (53% vs. 47%, respectively) [28]. Boutros performed a cross-sectional study of vegan and omnivorous diets for two consecutive years in healthy, young physically active women, which found that the vegan group had a significantly higher VO2 max (44.5 vs. 41.6 ml/kg/min) and submaximal endurance time to exhaustion (12.2 vs. 8.8 min) than omnivorous group despite no significant differences in percent body fat, BMI, physical activity level, muscle strength between groups [78]. Veleba recapitulated these findings in a randomized controlled trial, showing an increase in the vegetarian group's VO2max (12% increase), an increase which was not observed in the conventional diet group [74]. Durkalec-Michalski conducted a randomized and parallel intervention study of adult moderately-trained CrossFit participants for a 4-week high-intensity functional training study period and found significant improvements in squat repetitions and Wall Ball repetitions in the mixed diet group and additionally found significant improvements in deadlift, Wall Ball, and rowing repetitions in the vegetarian group, demonstrating improvements in both groups but inconsistency between groups [79].

Among runners, Nebl conducted a cross-sectional study of recreational runners, and found no significant differences in running distance, running time, training frequency, maximum power output, arterial lactate throughout exercise, and maximum lactate between plant-based, vegan, and non-plant-based dieting athletes [8]. Blancquaert similarly found no differences in performance measures (VO2 max, cycling time to exhaustion) between participants randomized to vegetarian and omnivorous diets for 6 months in their intervention study [47], and Page further noted no differences in VO2 max in cross-sectional study of young, healthy, active habitual vegans compared to habitual omnivores [76]. Several case studies offer similar results, following strictly vegan athletes who still perform extreme endurance-based competitions such as Triple Ironman events and cross mountainous races [80–83].

Discussion

The debate regarding the advantages and/or disadvantages of a plant-based diet has been ongoing for many years, and as plant-based diets become increasingly popular, there has been more query into the potential effect on athletic performance. With the progressive advances of technology, improved levels of communication and idea exchange, and the expansive spread of information through modern media, the interest in the relationship between diet, health, and performance has recently escalated.

Plant-based diets may offer the advantage of higher production of energy storage, which may enable athletes to perform at higher levels and for longer periods of time. As described by Williams et al. and Kersick et al., plant-based diets are associated with higher levels of carbohydrate intake, which may lead to increased glycogen stores and consequently a greater energy supply, thereby facilitating improved training adaptations [13, 14]. However, the theoretical advantages of greater antioxidant composition, elevated alkalinity, and reduced blood viscosity in a plant-based diet leading to reduced inflammation and improved immunity, attenuated muscular acidity and improved recovery, and greater oxygen delivery to tissues, respectively, have been shown to be inaccurate by several groups of researchers [7, 10, 15–18].

A restricted diet devoid of non-plant-based substances has been shown to lack components that may be important to overall health. As depicted by Fuhrman et al., plant-based dieters exhibit deficiencies in several micronutrients, which may lead to individual health issues [19]. Additionally, Raben et al. found that individuals who maintain plant-based diets have significantly lower fasting serum testosterone [9], which has been shown by Handelsman and colleagues to be associated with reduced muscle mass and strength [26]. Supplementation on an individual basis may help modulate these deficiencies, and it may reduce possible medical complications or performance insults secondary to the lack of critical nutrients.

Plant-based diets are on average less proteinaceous than non-plant-based diets, however, both diet styles can attain appropriate daily protein intake levels [28, 30], suggesting that adequate protein ingestion is not necessarily dependent solely on its source. Micronutrient intake, however, is dependent on source, as some micronutrients are derived strictly from animal-based products, which places plant-based dieters at higher risk of micronutrient deficiency if unaware [12, 19, 36, 37]. The organic compound creatine is found at higher levels in the blood of non-plant-based dieters when compared to plant-based-dieters on average, which ultimately may affect strength and power of athletes regardless of age; notwithstanding, creatine supplementation has been shown to be beneficial to plant-based athletes to compensate for power creatine levels [4, 20, 84]. These findings may imply that though diets contain varying levels of energy- and power-providing materials, ultimately these differences can be negated with supplemental intake or the differences may be clinically irrelevant. However, the concept of *food synergy*—which maintains that the biological constituents of nutrition are coordinated—would suggest that isolated nutritional supplementation is inferior to consumption of the whole food carriers of coordinated constituents [85, 86]. Jacobs et describe the "thinking food first"

approach, advocating for whole food ingestion of integrated constituents rather than consumption of isolated constituents created beyond the normal biological processes [87]. In circumstances where whole foods are unavailable to an athlete or incompatible with their dietary practices, then supplementation may be the preferred alternative.

Plant-based diets have been associated with lower average BMI, cholesterol levels, and blood pressure, as well as a lower incidence of T2DM, and reduced overall cancer and cardiovascular risk when compared to non-plant-based diets [27, 36, 37, 50–67]. There is evidence, however, to support that diets with high concentrations of meat and fish are associated with decreases in mean weight, BMI, waist circumference, systolic blood pressure, and fat composition as compared to pre-intervention baselines [68–71], which creates a conflict of arguments. Given that there are observable improvements identified across studies in the health outcomes yielded by both plant-based and non-plant-based diets, there is likely a presence of confounding contributors to the development of the described poor health outcomes (i.e. obesity, T2DM, malignancy, cardiovascular disease). Processed food intake is one such confounding factor, which has been shown to be associated with greater calorie, carbohydrate, and fat intake when compared to unprocessed food intake. Hall et al. conducted a randomized controlled trial, which found that ultra-processed diets led to weight gain whereas unprocessed diets led to weight loss [88]. Further confounders may not have yet been identified in the literature.

Despite the notable differences in the compositions of plant-based and omnivorous diets, multiple surrogate measures of strength and endurance showed no significant differences between athletes consuming plant-based or non-plant-based diets [2, 89]. Haub et al. and Lynch et al. found no significant differences in one-repetition maximum weight lifts and leg flexion/extension dynamometer exercises, respectively, between diet types [6, 28], and Baguet et al. showed no significant differences in power output while sprinting between athletes of each diet category [75]. In a separately reported study, Haub et al. found no differences in body composition, resting energy expenditure, and concentrations of muscle creatine/phosphocreatine/total creatine. Notably, the plant-based diets were supplemented with texturized vegetable proteins in several studies, which may have clouded the true strength and power performance of the plant-based dieters; additionally, these studies were conducted in elderly male participants, perhaps limiting the external validity of the results [6, 10].

There is mixed evidence regarding the effect of diet on performance during endurance events. Hietavala et al. and Lynch et al. found oxygen consumption to be higher during high intensity endurance exercise in plant-based athletes [7, 28]. On the other hand, Nebl et al. found no significant differences in running distance, running time, training frequency, maximum power output, arterial lactate throughout exercise, and maximum lactate during endurance activity [8], and Raben et al. found no differences between plant-based dieting athletes and non-selective dieting athletes when using maximal oxygen uptake, time to exhaustion, maximal voluntary contraction, and isometric endurance as surrogates for exercise endurance capacity [9]. In sum, there is no convincing or consistent data that plant-based diets and non-plant-based diets significantly differ in their outputs of strength and endurance.

Taken together, the findings of this clinical review support the recommendation for a nutritionally sufficient diet, rich in valuable contributing nutrients, energy, and proteins, regardless of source. The evidence does not support an advantage in performance based on strict adherence to a specified diet, but rather inclusion of the critical components that translate to performance, as there are high-performing athletes on the elite stage in a variety of competitions that practice both plant-based and non-selective dieting lifestyles [82, 83].

The findings in this review should be interpreted considering several limitations. This is a clinical review of a limited number of studies about health and performance as it relates to diet. These studies vary in sample size, study design, outcome metrics, and patient selection and therefore have inherent biases. This review sheds light on an opportunity to pursue further investigation with the goal to limit bias and identify causative relationships. Specifically, while RCTs are expensive and can be intrusive, this line of research is necessary to make definitive conclusions about the impact of plant-based diets on various aspects of athletic performance. Future research may focus on the perspectives of athletes, with a particular focus on the perceived contributions of diet types on nutrition and performance, and possible collaboration with the ongoing efforts into understanding the gut microbiome and metagenomics as to how they connect diet, nutrition, and performance [90, 91]. Additionally, investigations into performance differences based on diet variations may be pursued in professional athletes, where the environment may be controlled to reduce variables interfering with the effects of diet on performance. Furthermore, many of the studies investigating diet and performance are limited by a short study period often 6–12 weeks; it is possible that nutritional homeostasis may take longer to achieve, and thus longer-term investigations may elucidate relationships that are unable to develop in shorter periods of time.

Conclusion

As plant-based diets have become increasingly popular, more research has investigated the effect of plant-based diets as it relates to athletic performance. Plant-based diets

have been shown to offer improved physiologic benefits, including improved cardiovascular performance, lower rates of T2DM, lower rates of malignancy [27, 36, 37, 43, 53, 54, 59, 60, 63, 65, 92, 93]. These health benefits, however, have not been shown to translate directly into improved athletic performance. Some common concerns with plant-based diets relate to lower creatine levels and micronutrient deficiencies, leading to theoretical submaximal performance. However, these changes have not been shown to translate to reduced athletic performance. Though plant-based and mixed omnivorous eaters vary in their macronutrient and micronutrient intake, disease propensities, and oxygen consumption during exertion, a plant-based diet neither significantly hinders nor improves physical performance across measures of strength, power, and aerobic/anaerobic performance. This lays credence to the fact that plant-based diets offer satisfactory caloric and nutritional value, even for high performing athletes. For athletes, trainers, nutritionists and dieticians, and physicians, the most important recommendation is to maintain a nutritionally sufficient diet that provides the appropriate levels of vital nutrients and vitamins, with supplementation as needed.

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Declaration

Conflict of interest The authors did not receive support from any organization for the submitted work. All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

Ethical approval and Informed consent This study was conducted within the agreed upon standards of expected ethical behavior. As there was no involvement of human subjects, consent was not needed to proceed with the investigation.

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